DEMO

prepare a prototype (flow + structure) for a **Document Management System (DMS)** based on your detailed workflow (Creator → Approver → Issuer → Controlled Access).

-System Flow Chart (process steps)

-Module-wise Functionality (dashboard, upload, approval, issuance, access control)

-Database Design (sample schema)

-User Role Permissions

-Technology suggestion for development

Creator Login

Fill Doc Info (a–g fields)

Upload Word Doc (No header/footer)

Send to Approver (Notify)

Approver reviews doc (with header/footer)

Approve Document Decline (add remark)

Add remarks, Goes back to Creator

select sharing dept.

Send to Issuer (MR)

Issuer reviews doc (header/footer)

Issue Final Copy Decline (add remark)

Add Issue remark Goes back to Creator

Convert to PDF (Final)

Auto Notify All Users

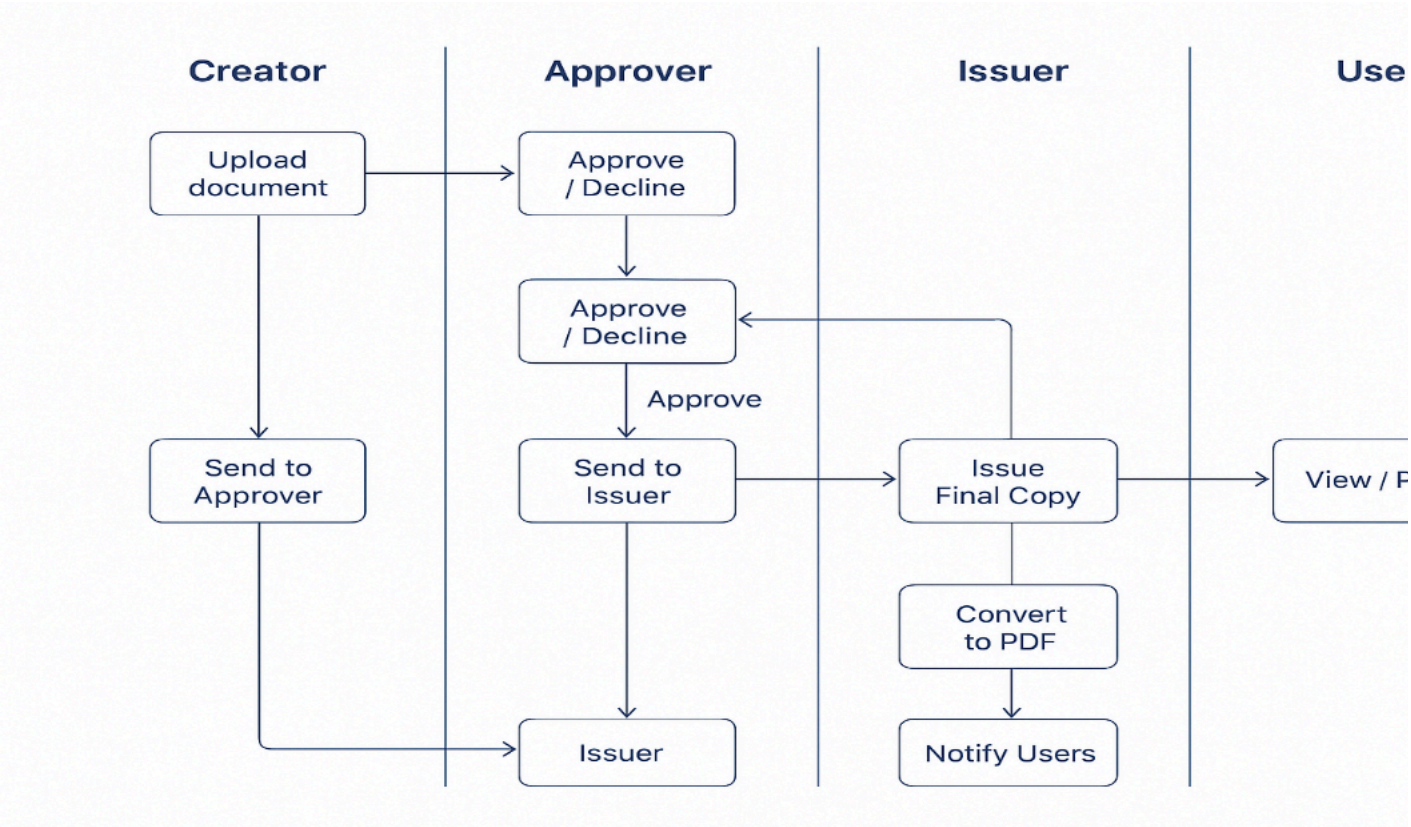
Assign Control Copy Nos

Users View/Print Controlled PDF only

(No download, traceable by User ID)

Responsibility: Module-Wise Functional Prototype Plan

Role	Features
Creator	Create new document entry, upload Word doc, view pending/returned docs, resubmit
Approver	View pending approvals, open doc (auto header/footer), approve/reject, add remarks, assign departments
Issuer (MR)	View approved docs, issue/reject, add final remark, track issued documents
User/Recipient	View issued documents (PDF only), print with control footer
Admin	Manage users, roles, departments, track logs, export reports



There is some conditions:

-Auto reminders for revision (based on due period)

-Comparison tool (old vs. revised doc)

-Version history tracking

-Export to Excel logs (per user or document)

-QR code footer for authenticity

About Database table need:

users

Column	Type	Description
user_id	INT (PK)	Unique user
name	VARCHAR	User name
role	ENUM('Creator','Approver','Issuer','User','Admin')	Role
department	VARCHAR	Dept name
email	VARCHAR	Notification use

documents Column	Type	Description
doc_id	INT (PK)	Unique document
doc_name	VARCHAR	SOP/Format name
doc_number	VARCHAR	Unique doc code
date_of_issue	DATE	Issue date
revision_no	INT	Revision count
due_period_years	INT	Review reminder

documents **Column**	**Type**	**Description**
prepared_by	INT (FK)	Creator ID
reason_revision	TEXT	Revision reason
file_path	VARCHAR	Uploaded file
status	ENUM('Pending','Approved','Declined','Issued')	Current status
approver_id	INT (FK)	Assigned approver
issuer_id	INT (FK)	Assigned issuer

document_log

Column	**Type**	**Description**
log_id	INT (PK)	Log entry
doc_id	INT (FK)	Related document
user_id	INT (FK)	Who performed action
action	VARCHAR	Created, Approved, Declined, Issued
remarks	TEXT	Action remarks
date_time	DATETIME	Timestamp

document_access **Column**	**Type**	**Description**
doc_id	INT (FK)	Document
user_id	INT (FK)	Recipient
control_copy_no	VARCHAR	Auto serial copy no.
view_access	BOOLEAN	Can view PDF
print_access	BOOLEAN	Can print PDF
print_date	DATETIME	Track printed date

Feature	Creator	Approver	Issuer	Recipient	Admin
Create Doc	✅	❌	❌	❌	✅
Approve Doc	❌	✅	❌	❌	✅
Issue Doc	❌	❌	✅	❌	✅
View PDF	✅ (Issued only)	✅	✅	✅	✅
View Previous Revisions	❌	❌	✅ (Master copy)	❌	✅
Export Logs	✅	✅	✅	❌	✅

Layer	Technology
Frontend (Web UI)	React.js / Next.js (with Role-based Dashboard)
Backend API	Node.js (Express) or ASP.NET Core
Database	PostgreSQL / SQL Server
File Storage	Local folder or AWS S3 / Supabase Storage
Notifications	Email + In-app popup via Socket.io or Supabase Realtime
Document Conversion	LibreOffice / Word API → PDF (with watermark + control footer)
Audit Logging	Automatic via middleware (store all user actions)